DeepSeek-V2: A Strong, Economical, and Efficient MoE Language Model

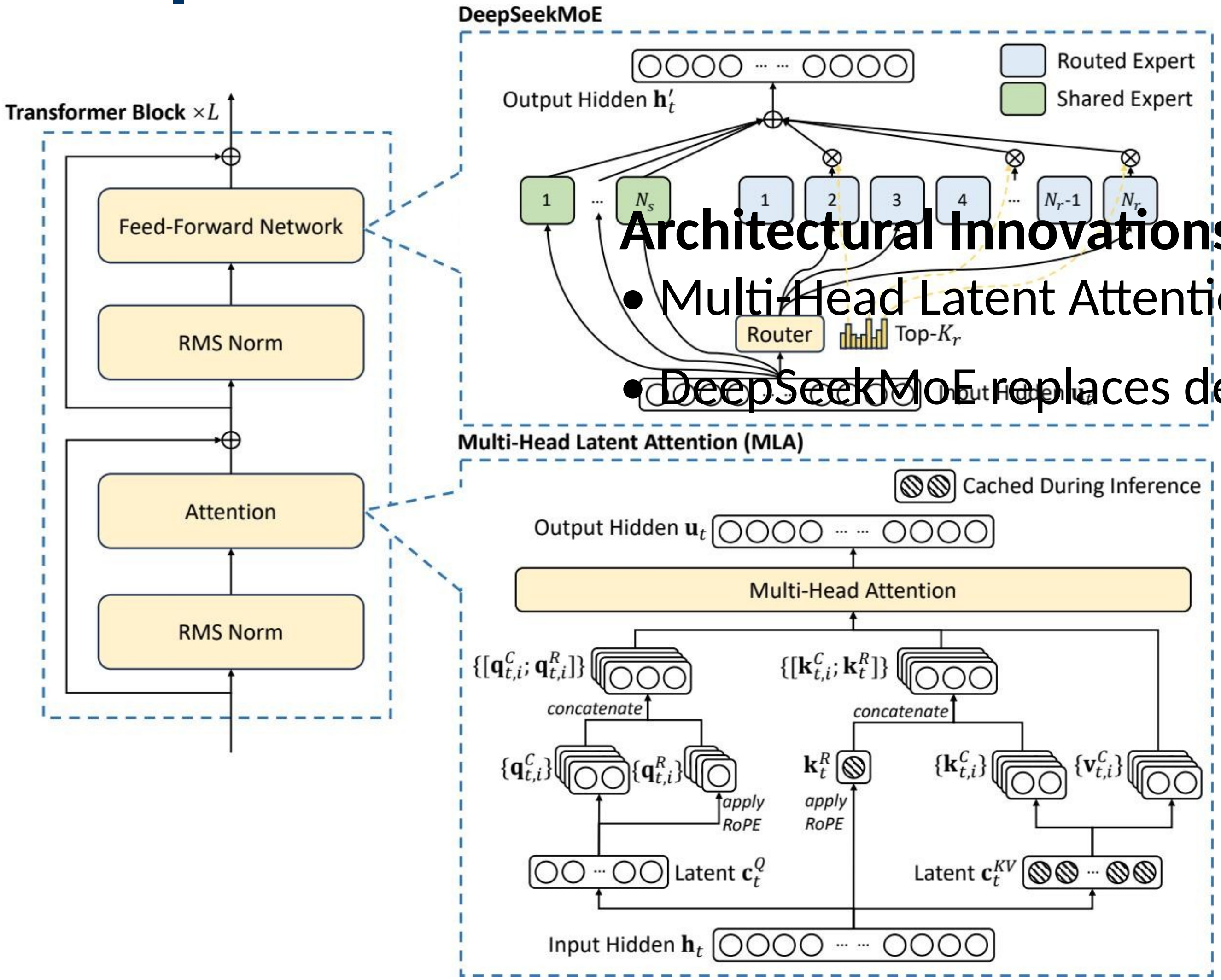
236B Total Parameters | 21B Activated | 128K Context

Based on: DeepSeek-V2: Efficient MoE Language Model with Multi-Head Latent Attention

The LLM Scaling Dilemma: Performance vs. Training Cost and Inference Speed

- Scaling up model parameters improves reasoning and intelligence but incurs prohibitive training costs.
- Conventional dense models suffer from slow inference due to Key-Value (KV) cache bottlenecks.
- As context length increases (e.g., 128K tokens), the KV cache consumes massive memory, limiting concurrent generation throughput.
- A new approach is required: Mixture-of-Experts (MoE) for efficient training, coupled with an innovative attention mechanism to drastically reduce KV cache.

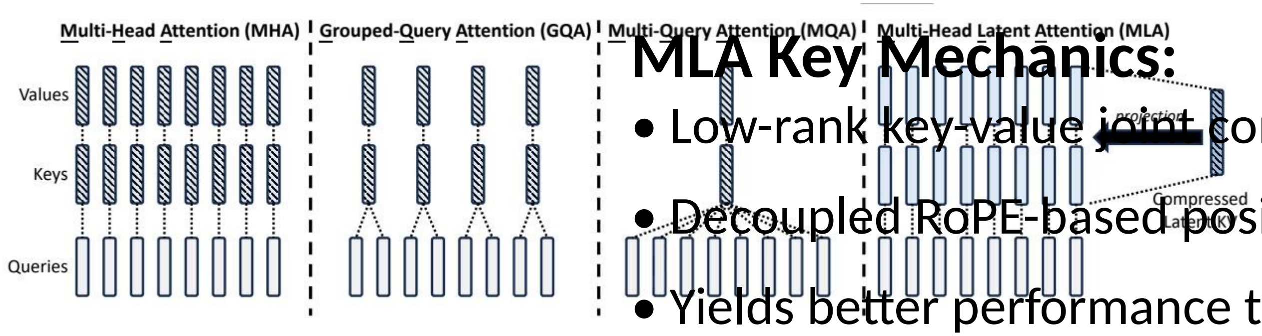
DeepSeek-V2 Architecture: MLA + DeepSeekMoE in a Transformer Block



Architectural Innovations:

- Multi-Head Latent Attention (MLA) replaces standard Multi-Head Atten
- DeepSeekMoE replaces dense Feed-Forward Networks (FFN) within ea

Multi-Head Latent Attention Compresses KV Cache by 93.3%



MLA Key Mechanics:

- Low-rank key-value joint compression: Compresses K and V into a compact latent ve
- Decoupled RoPE-based positional key is reconstructed on-the-fly via learned projec
- Yields better performance than MHA while requiring dramatically less cache than G

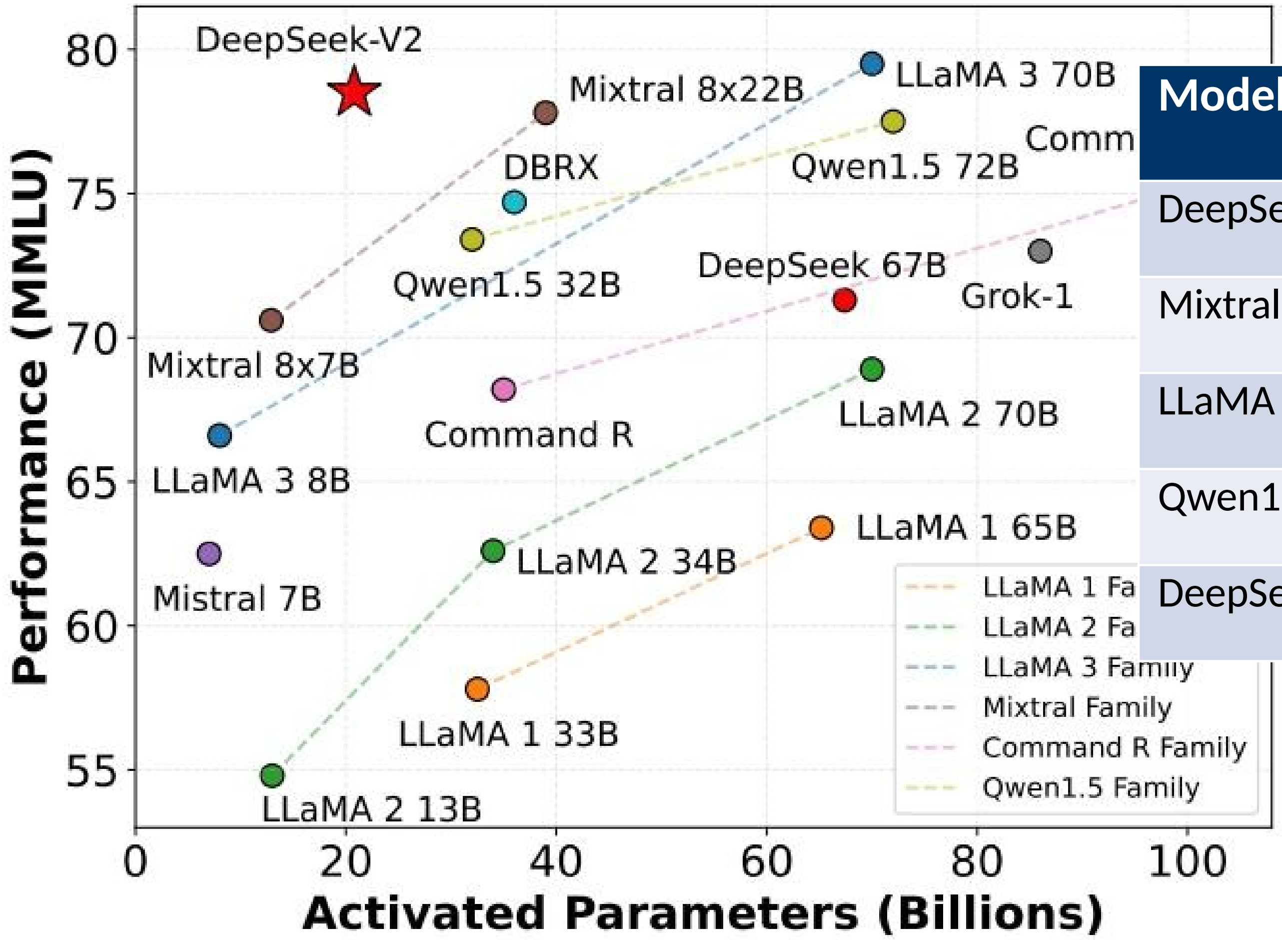
DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

- Shared Experts (Ns): Always activated to capture common knowledge across contexts.
- Routed Experts (Nr): Sparsely activated via a top-Kr router to ensure specialized knowledge processing.
- Fine-grained expert segmentation increases the potential for specialization.
- Device-limited routing constrains each token to experts on at most M devices, reducing cross-node communication overhead.
- Auxiliary losses enforce load balance across experts and devices.
- Token-dropping strategy during training prevents expert overflow without degrading inference quality.

42.5% Training Cost Reduction and 5.76× Generation Throughput vs. DeepSeek 67B

Metric	DeepSeek 67B (Baseline)	DeepSeek-V2 (MoE + MLA)
Architecture	Dense MHA	MoE with Multi-Head Latent Attention
Training Cost	Baseline (100%)	42.5% Reduction
KV Cache Size	Baseline (100%)	93.3% Reduction
Max Generation Throughput	Baseline (1.00×)	5.76× Improvement

DeepSeek-V2 Tops MMLU Among Open-Source Models at Only 21B Activated Parameters



Model	Activated Params (B)	MMLU Score
DeepSeek-V2	21	78.5
Mixtral 8x22B	39	77.6
LLaMA 3 70B	70	77.0
Qwen1.5 72B	72	75.6
DeepSeek 67B	67	73.0

DeepSeek-V2 Chat (RL) Rivals Closed-Source Models on Chinese Benchmarks

Alignment Pipeline:

- Supervised Fine-Tuning (SFT) on 1.5M conversational sessions.
- Reinforcement Learning via Group Relative Policy Optimization (GRPO).
- Outperforms all open-source and most closed-source models in Chinese open-ended evaluation.

Benchmark	DeepSeek-V2 Chat (RL) Score
AlpacaEval 2.0 (length-controlled win rate)	38.9%
MT-Bench (overall)	8.97
AlignBench (overall, Chinese)	7.91

Key Takeaways: Architectural Innovation Unlocks Efficient Scaling

- MLA is a strictly superior KV-cache solution, compressing KV cache by 93.3% while exceeding MHA performance.
- DeepSeekMoE enables highly economical training (42.5% lower cost) through fine-grained experts and shared expert isolation.
- DeepSeek-V2 establishes a new Pareto-optimal frontier, achieving top-tier MMLU performance using only 21B activated parameters.
- Strong multilingual alignment pipeline positions DeepSeek-V2 Chat as the leading open-source model in Chinese.
- 5.76× higher generation throughput ensures the model is highly practical for large-scale production deployment